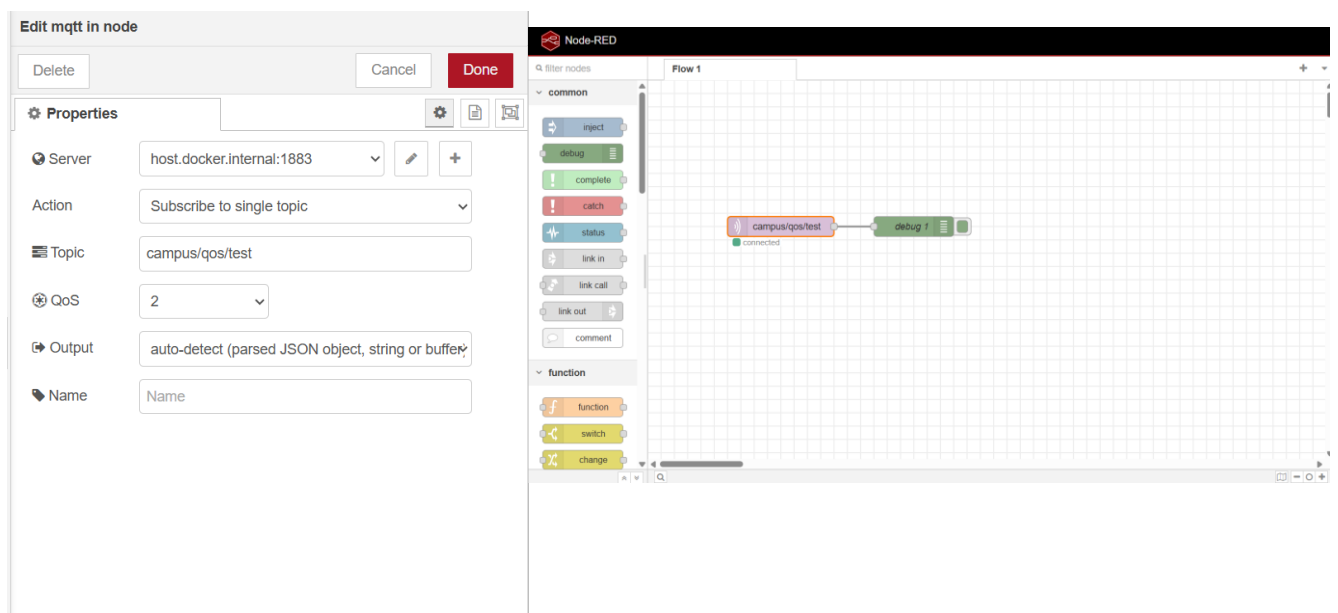


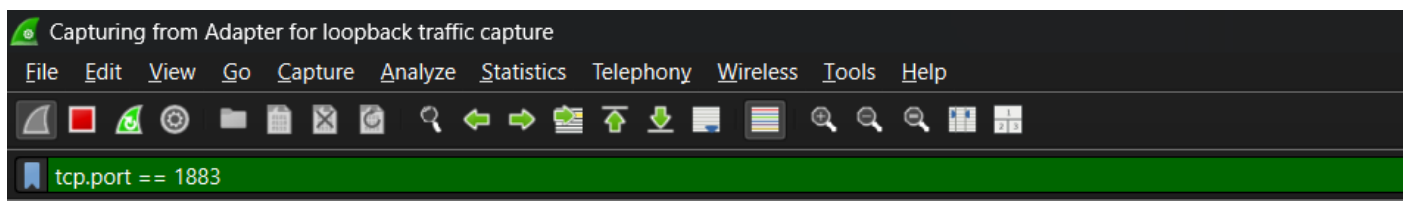
Exercise 5: MQTT Reliability (QoS Levels, Handshake & Retained Messages)

The objective of this exercise is to understand MQTT communication between publisher and subscriber using Node-RED and Postman, to analyse Quality of Service (QoS) levels (0, 1, and 2), and to observe MQTT packet behaviour using Wireshark. This includes studying message reliability mechanisms and retained message functionality in an IoT environment.

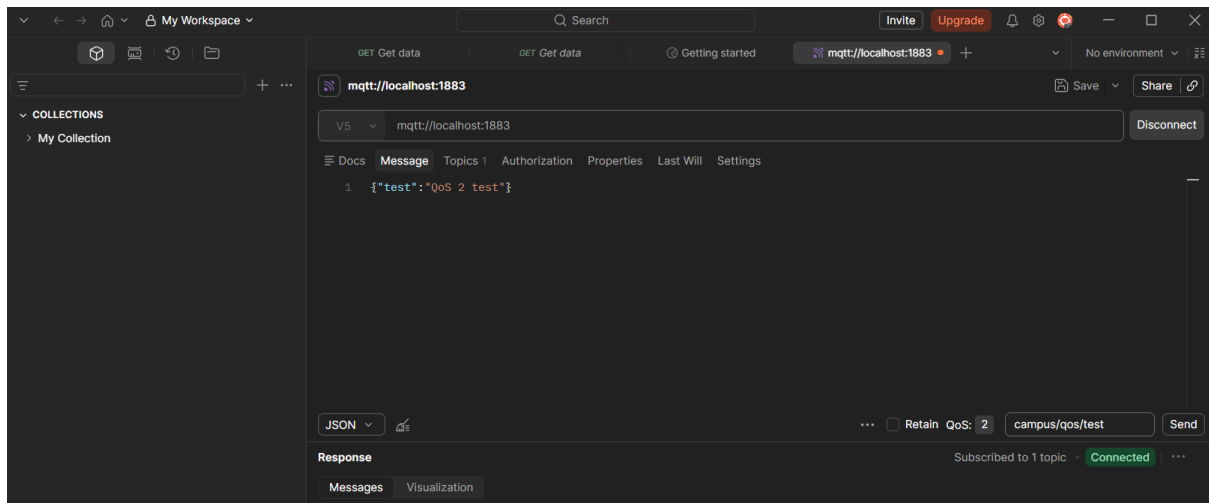
STEP 1 :SETUP NODE RED



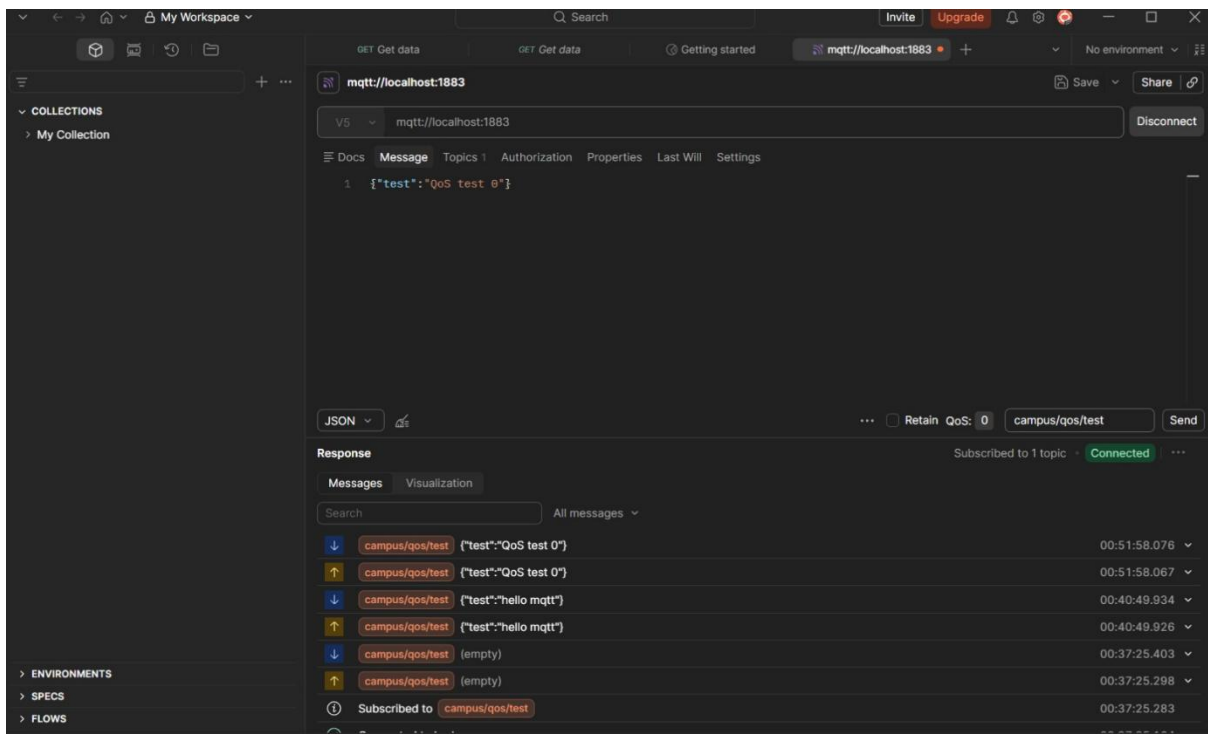
STEP 2 :SETUP WIRESHARK



STEP 3 :SETUP POSTMAN



AT QOS 0



Capturing from Adapter for loopback traffic capture

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tcp.port == 1883

No.	Time	Source	Destination	Protocol	Length	Info
17115	3159.0285545...	127.0.0.1	127.0.0.1	TCP	56	[TCP Keep-Alive ACK] 49672 → 1883 [ACK] Seq=68 Ack=58 Win=65280 Len=0 SLE=57 SRE=58
17116	3159.0285601...	127.0.0.1	127.0.0.1	TCP	45	[TCP Keep-Alive] 49672 → 1883 [ACK] Seq=67 Ack=58 Win=65280 Len=1
17117	3159.0285721...	127.0.0.1	127.0.0.1	TCP	56	[TCP Keep-Alive ACK] 1883 → 49672 [ACK] Seq=58 Ack=68 Win=65280 Len=0 SLE=67 SRE=68
17140	3160.980446...	:::1	:::1	MQTT	105	Publish Message [campus/qos/test]
17141	3160.9804804...	:::1	:::1	TCP	64	1883 → 54169 [ACK] Seq=165 Ack=247 Win=65280 Len=0
17142	3160.9814774...	:::1	:::1	MQTT	105	Publish Message [campus/qos/test]
17143	3160.9814991...	:::1	:::1	TCP	64	54169 → 1883 [ACK] Seq=247 Ack=206 Win=65280 Len=0
17144	3160.9816178...	127.0.0.1	127.0.0.1	MQTT	84	Publish Message [campus/qos/test]
17145	3160.9816371...	127.0.0.1	127.0.0.1	TCP	44	49672 → 1883 [ACK] Seq=68 Ack=98 Win=65280 Len=0
17228	3172.4927715...	:::1	:::1	MQTT	105	Publish Message [campus/qos/test]
17229	3172.4928126...	:::1	:::1	TCP	64	1883 → 54169 [ACK] Seq=206 Ack=288 Win=65024 Len=0
17230	3172.4937882...	:::1	:::1	MQTT	105	Publish Message [campus/qos/test]
17231	3172.4938144...	:::1	:::1	TCP	64	54169 → 1883 [ACK] Seq=288 Ack=217 Win=65280 Len=0
17232	3172.4939706...	127.0.0.1	127.0.0.1	MQTT	84	Publish Message [campus/qos/test]
17233	3172.4939925...	127.0.0.1	127.0.0.1	TCP	44	49672 → 1883 [ACK] Seq=68 Ack=138 Win=65280 Len=0
17242	3173.9990001...	127.0.0.1	127.0.0.1	MQTT	46	Ping Request
17243	3173.9990266...	127.0.0.1	127.0.0.1	TCP	44	1883 → 49672 [ACK] Seq=138 Ack=70 Win=65280 Len=0
17244	3173.9997895...	127.0.0.1	127.0.0.1	MQTT	46	Ping Response
17245	3173.9998048...	127.0.0.1	127.0.0.1	TCP	44	49672 → 1883 [ACK] Seq=70 Ack=140 Win=65280 Len=0

Frame 3063: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF_{...}

Null/loopback

Internet Protocol Version 6, Src: ::1, Dst: ::1

Transmission Control Protocol, Src Port: 60669, Dst Port: 1883, Seq: 0, Len: 0

0000 18 00 00 00 60 05 f1 1a 00 20 06 80 00 00 00 00 ...

0010 00 00 00 00 00 00 00 00 00 00 01 ec fd 07 5b ...

0020 00 00 00 00 00 00 00 00 00 00 01 ec fd 07 5b ...

0030 10 fb 50 4e 00 00 00 00 80 02 ff ff 1f 5d 00 00 ...

0040 02 04 ff c3 01 03 08 01 01 04 02 ...

wireshark: Adapter for loopback traffic capture\OLAR3.pcapng

Packets: 17321 - Displayed: 236 (1.4%)

Profile: Default

Node-RED

Flow 1

campus/qos/test → debug 1

debug

6/24/2026, 12:51:58 AM node: debug 1

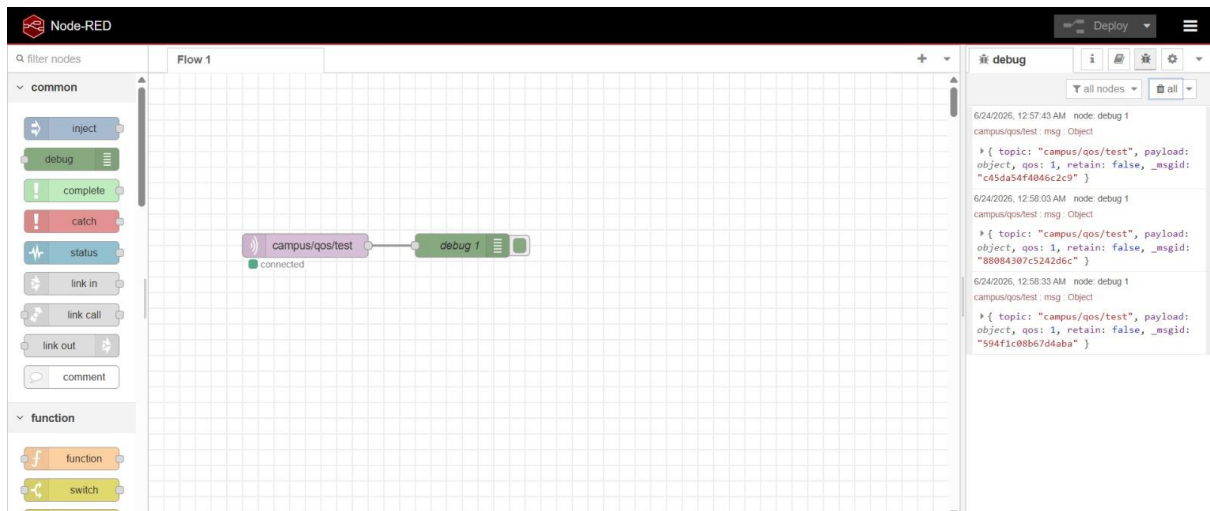
campus/qos/test: msg: Object

{ topic: "campus/qos/test", payload: object, qos: 0, retain: false, _msgid: "b3fcd5388643fb9d" }

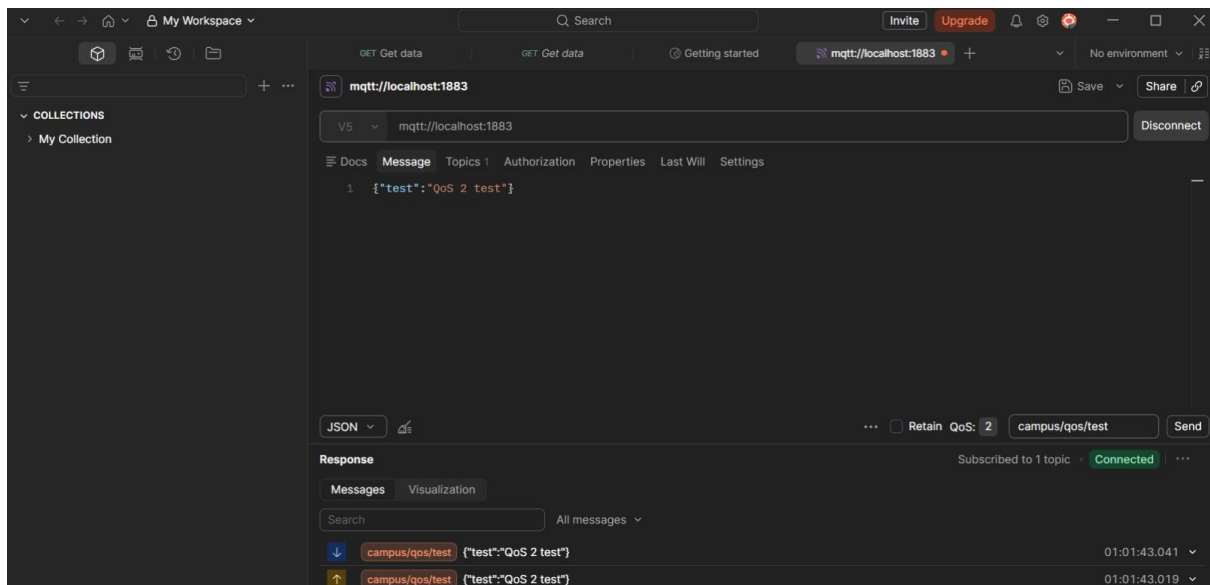
AT QoS 1

The screenshot shows the MQTT Explorer application interface. At the top, there's a search bar and a 'My Workspace' dropdown. Below that, a sidebar on the left shows 'COLLECTIONS' and 'My Collection'. The main area displays a message from 'mqtt://localhost:1883' with the payload `{\"test\":\"QoS 1 test\"}`. The message is shown in JSON format. Below the message, there's a 'Response' section with 'Messages' and 'Visualization' tabs. The 'Messages' tab is active, showing a search bar and a list of messages. The first message is selected, showing its details: 'campus/qos/test' with a QoS of 1. The status bar at the bottom indicates '00:58:33.178'.

The screenshot shows a Wireshark network traffic capture. The top bar indicates 'Capturing from Adapter for loopback traffic capture'. The main pane shows a list of captured packets. The selected packet is a MQTT Publish Message (id=2) with a QoS of 1, sent from 127.0.0.1 to 127.0.0.1. The packet details pane shows the MQTT message structure, including the Topic 'campus/qos/test' and the Payload `{\"test\":\"QoS 1 test\"}`. The packet bytes pane shows the raw data of the message, including the MQTT header and the JSON payload.



AT QOS 2



Capturing from Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 1883

No.	Time	Source	Destination	Protocol	Length	Info
20365	3582.9315927...	:::1	:::1	TCP	64	54169 → 1883 [ACK] Seq=429 Ack=400 Win=65024 Len=0
20446	3592.7413498...	:::1	:::1	MQTT	107	Publish Message (id=48475) [campus/qos/test]
20447	3592.7413827...	:::1	:::1	TCP	64	1883 → 54169 [ACK] Seq=480 Ack=472 Win=65024 Len=0
20448	3592.7435550...	:::1	:::1	MQTT	70	Publish Received (id=48475)
20449	3592.7436065...	:::1	:::1	TCP	64	54169 → 1883 [ACK] Seq=472 Ack=406 Win=65024 Len=0
20450	3592.7437240...	:::1	:::1	MQTT	105	Publish Message [campus/qos/test]
20451	3592.7437430...	:::1	:::1	TCP	64	54169 → 1883 [ACK] Seq=472 Ack=447 Win=65024 Len=0
20452	3592.7448614...	:::1	:::1	MQTT	70	Publish Release (id=48475)
20453	3592.7449090...	:::1	:::1	TCP	64	1883 → 54169 [ACK] Seq=447 Ack=478 Win=65024 Len=0
20454	3592.7456155...	:::1	:::1	MQTT	70	Publish Complete (id=48475)
20455	3592.7456413...	:::1	:::1	TCP	64	54169 → 1883 [ACK] Seq=478 Ack=453 Win=65024 Len=0
20456	3592.7472465...	127.0.0.1	127.0.0.1	MQTT	86	Publish Message (id=4) [campus/qos/test]
20457	3592.7472874...	127.0.0.1	127.0.0.1	TCP	44	49672 → 1883 [ACK] Seq=94 Ack=320 Win=65024 Len=0
20458	3592.7489905...	127.0.0.1	127.0.0.1	MQTT	48	Publish Received (id=4)
20459	3592.7490272...	127.0.0.1	127.0.0.1	TCP	44	1883 → 49672 [ACK] Seq=320 Ack=98 Win=65280 Len=0
20460	3592.7497917...	127.0.0.1	127.0.0.1	MQTT	48	Publish Release (id=4)
20461	3592.7498196...	127.0.0.1	127.0.0.1	TCP	44	49672 → 1883 [ACK] Seq=98 Ack=324 Win=65024 Len=0
20462	3592.7521713...	127.0.0.1	127.0.0.1	MQTT	48	Publish Complete (id=4)
20463	3592.7522086...	127.0.0.1	127.0.0.1	TCP	44	1883 → 49672 [ACK] Seq=324 Ack=102 Win=65280 Len=0

Frame 18737: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF...
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 1883, Dst Port: 49672, Seq: 188, Ack: 80, Len: 42
MQ Telemetry Transport Protocol, Publish Message

wireshark_Adapter for loopback traffic capture\OLAR3.pcapng

Packets: 20545 - Displayed: 410 (2.0%)

Profile: Default

Node-RED

Flow 1

campus/qos/test → debug 1

debug

6/24/2026, 1:01:43AM node debug 1
campus/qos/test :msg: Object
[{"topic": "campus/qos/test", "payload": {"object": "qos: 2", "retain": false, "_msgid": "a4b5e3b87a51cc45"} }]